

Supply Chain Digital Twin using Reinforcement Learning

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This is to certify that the project titled “*Supply Chain Digital Twin using Reinforcement Learning*”, submitted by Ayaan Khan, Arya Patil, Nihaal Badam, and Swastik Ankulge in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering, VNIT Nagpur. The work is comprehensive, complete and fit for final evaluation.

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Abstract

Supply chain management in urban logistics networks faces a fundamental challenge: Delivery fleets must satisfy multiple competing objectives simultaneously, including minimizing fuel costs, preserving cargo freshness, maximizing delivery speed, and avoiding unsafe road conditions. Traditional route planning algorithms as A* uses a single fixed cost function and cannot adapt dynamically to real-time events like road accidents, extreme weather, or sudden demand spikes. This project addresses this limitation by designing and implementing a Supply Chain Digital Twin for Nagpur, validated over a 365-day simulation.

The system is built around a three-tier AI architecture. Central PPO (Proximal Policy Optimization) Brain, trained for 50,000 steps to select macro-level routing strategies from three options: Fuel-Priority, RSL-Priority, and Speed-Priority. Next is a per-truck Edge Q-Learning Agent that performs real-time emergency micro-rerouting when accidents or weather events block the planned route. The last is an online Ridge Regression ETA Forecaster that predicts segment travel times using an 8-dimensional spatial-temporal feature vector and updates itself continuously throughout the simulation.

The system was evaluated through a 365-day comparative benchmark against a heuristic A* baseline. The AI-enabled configuration fulfilled 323 additional orders, delivered 474,708 more kilograms of cargo, and reduced retailer stockout events from 70,898 to 33,401, a reduction of 52.9%. Road accidents were reduced by 135 events as an emergent consequence of the ETA Forecaster learning to route trucks away from high-risk zones during peak hours. Cargo freshness at delivery and fleet fuel efficiency were identical in both configurations, at 99.7% and 78.7% respectively, confirming that the throughput gains were achieved without any trade-off in quality or operational cost. The results demonstrate Pareto optimality: six of eight performance metrics improved simultaneously, and no metric regressed.

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Chapter 1

Introduction

This chapter provides an overview of the problem addressed in this project. Section 1.1 introduces the supply chain management domain and discusses the background, the problem being solved, the proposed solution, and the expected outcomes. Section 1.2 lists the specific technical contributions made through this work. Section 1.3 outlines how the remainder of the thesis is organized.

1.1 Supply Chain Management

1.1.1 Background

The supply chain is the sequence of processes through which raw materials are sourced, processed, and delivered as finished goods to end consumers. In the context of fast-moving consumer goods (FMCG), this chain typically spans manufacturers, regional distribution hubs, city-level warehouses, and retail outlets. Effective management of this chain directly affects product availability, operational costs, and customer satisfaction. Traditional supply chain management (SCM) relied primarily on deterministic models, fixed routing schedules, and rule-based reorder policies. These approaches performed adequately when demand was stable and supply was predictable.

India presents a particularly challenging environment for supply chain management. The rapid growth of e-commerce, the expansion of cold-chain logistics for perishable goods, and the diversity of urban road networks have made static routing policies increasingly inadequate. Tier-2 cities such as Nagpur represent a critical and growing segment of this logistics challenge. Nagpur serves as a major road and rail junction in central India, making it a significant node in distribution networks that cover Maharash-

tra and the surrounding states. Its road network spans multiple distinct zones, including residential neighborhoods with high residential traffic density, commercial office corridors, shopping and market districts, and high-speed highway bypasses. Each of these zones exhibits a different traffic profile depending on the time of day and the day of the week, making static routing ineffective even within a single city.

The problem is compounded when cargo has a limited shelf life. Perishable goods such as dairy products, medicines, and fresh produce carry a Remaining Shelf Life (RSL) that decreases with every hour of transit. A truck that takes a slightly longer but less congested route may arrive on time, while a truck that follows the shortest path through a congested market zone may arrive with cargo that has deteriorated significantly. Simultaneously, fleet operators face pressure to minimize fuel consumption, since fuel costs represent a major portion of logistics expenses. These competing constraints, namely freshness, speed, fuel efficiency, and road safety, cannot be balanced simultaneously by any single static rule. This motivates the use of artificial intelligence to make adaptive, real-time routing decisions.

1.1.2 Problem Statement

The core problem addressed in this project is the dynamic routing of a delivery fleet under multiple competing real-time constraints in an urban logistics network. The specific challenges are as follows.

First, delivery routes must respond to conditions that change continuously: road accidents, unexpected weather events such as monsoon rainfall, and fluctuating traffic density across different zones and times of day. A route that is optimal at 9 AM may be severely suboptimal by 11 AM once office-hour congestion sets in.

Second, cargo freshness imposes a time-based deadline on each delivery. Once a batch of goods is loaded onto a truck, the RSL begins declining. The routing system must factor in how much freshness a cargo will have remaining when it reaches the retailer, not just how quickly the truck will physically arrive.

Third, fuel consumption is a direct cost. Routing the entire fleet aggressively for speed maximizes throughput but depletes fuel faster, increases refueling frequency, and raises operational costs. A purely fuel-conservative strategy, on the other hand, results in slower deliveries and more stockouts at retailer outlets.

Fourth, current heuristic routing algorithms such as A* use a single fixed cost function and cannot switch between optimization objectives based on the current state of the fleet. A truck that is almost out of fuel should be routed conservatively, while a truck carrying cargo with a short remaining shelf life should be routed aggressively for speed. No single static heuristic can capture this context-sensitive decision-making requirement.

1.1.3 Solution

This project proposes a Supply Chain Digital Twin for the city of Nagpur, implemented as a discrete-event simulation environment in which a fleet of delivery trucks operates over a realistic road network, subject to dynamic weather, traffic, and accident events. The Digital Twin serves as a safe, repeatable environment for training and validating artificial intelligence models before deployment.

The routing intelligence of the system is organized into a three-tier hierarchy. The first tier is a Central PPO Brain, a Proximal Policy Optimization (PPO) agent trained offline to observe the state of the full fleet and select a macro-level routing strategy from three options: Fuel-Priority, RSL-Priority, or Speed-Priority. The second tier is an Edge Q-Learning Agent embedded within each individual truck, which performs micro-level emergency rerouting in real time when sudden events such as accidents or extreme weather block the current route. The third tier is an ETA Forecaster based on online Ridge Regression, which predicts travel time for individual road segments based on real-time conditions including the zone, time of day, day of the week, weather, and traffic density. This prediction feeds directly into the route planner's cost function, improving the accuracy of route selection.

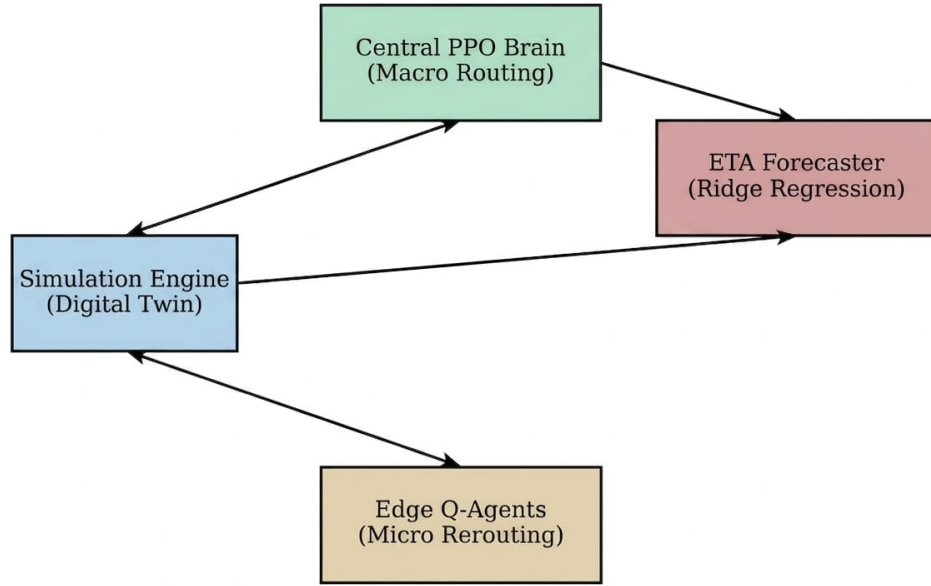


Figure 1.1: High-Level System Architecture of the Supply Chain Digital Twin

Figure 1.1 shows how these components interact. Orders are placed by retailer nodes and processed by the simulation engine. The Central PPO Brain observes the 25-dimensional fleet state and selects a strategy. The ETA Forecaster continuously refines its travel time estimates. The Edge Agents handle unexpected disruptions. Together, these three tiers enable the fleet to make intelligent, context-sensitive routing decisions across the full year of operation.

1.1.4 Outcomes

The system was validated through a 365-day comparative simulation benchmark, with both the baseline heuristic router and the full AI-enabled system run under identical conditions. The following outcomes were observed from the results.

- The AI-enabled system fulfilled 323 additional orders compared to the baseline over the full year, an increase of 4.7%.
- Total cargo delivered increased by 474,707 kg, representing nearly half a million additional kilograms of goods reaching retailers.

- Retailer stockout events were reduced from 70,898 to 33,401, a reduction of 52.9%, demonstrating substantially improved supply continuity.
- Road accidents were reduced from 2,212 to 2,077, showing that the AI learned to prefer safer routing patterns without being explicitly instructed to.
- Cargo freshness at the point of delivery averaged 99.7% in both the baseline and RL configurations, confirming that RSL was preserved throughout transit and not sacrificed for throughput.
- Fuel efficiency remained identical between the two configurations at 78.7%, proving that the throughput gains were achieved without any additional fuel cost.

1.2 Contribution

The specific technical contributions of this project are listed below.

- **Supply Chain Digital Twin Simulation Engine:** A complete discrete-event simulation of the Nagpur city logistics network was designed and implemented. The simulation models a multi-zone road graph, a fleet of delivery trucks, warehouse and retailer nodes, and dynamic environmental events including weather, traffic density, and road accidents. This engine serves as the training and evaluation environment for all AI components.
- **Central PPO Routing Agent:** A Proximal Policy Optimization agent was designed, trained for 50,000 steps, and deployed for macro-level strategy selection. The agent observes a 25-dimensional state vector and selects one of three routing strategies per dispatch cycle.
- **Edge Q-Learning Emergency Override Agent:** A per-truck tabular Q-Learning agent was designed for real-time emergency micro-rerouting. The agent operates entirely online, updating its Q-table after each emergency event, and can override the Central Brain's strategy when local road conditions make it dangerous or infeasible.

- **8-Dimensional Spatial-Temporal ETA Forecaster:** An online Ridge Regression model was designed and integrated to predict travel time for each road segment. The feature vector was upgraded from a 5-dimensional baseline to an 8-dimensional vector incorporating circular time-of-day encoding, circular day-of-week encoding, and geographic zone classification. This allows the forecaster to capture temporal and spatial traffic patterns simultaneously.
- **Reward Normalization Framework (5:1 Ratio):** A mathematical analysis of the physical magnitudes of fuel consumption and cargo decay rates was used to derive a 5-to-1 normalization ratio for the PPO reward function. This normalization prevents the PPO agent from defaulting to a single objective and enables true dynamic strategy switching.
- **Anti-Thrashing GPS State Lock:** A route-thrashing bug was identified and resolved through the implementation of a state machine lock inside the truck agent. This fix was critical: without it, the fleet delivery rate collapsed to 49.8% due to constant route recalculation.
- **365-Day Long-Term Validation Benchmark:** The system was validated over a full simulated year (365 days) under a fixed reproducible seed, producing a statistically meaningful comparison between the baseline heuristic router and the full AI-enabled system.

1.3 Organization of Thesis

This thesis is organized to systematically cover all aspects of the proposed system, from motivation through implementation and results.

Chapter 1 introduces the problem, the solution approach, and the specific technical contributions. Chapter 2 reviews the existing literature on traditional routing methods, machine learning in logistics, and the theoretical foundations of reinforcement learning. Chapter 3 describes the data generated by the simulation environment and how

it is preprocessed into state representations for the AI models. Chapter 4 presents the full design and implementation of all system components, including the simulation engine, the three-tier AI architecture, and the training procedure. Chapter 5 presents and analyzes the results from the 365-day benchmark. Chapter 6 concludes the thesis and discusses directions for future work. References are listed in Chapter 7 in IEEE numbered format.

Chapter Summary: This chapter introduced the problem of dynamic fleet routing under competing real-time constraints in the Nagpur city supply chain. The proposed solution, a three-tier AI architecture embedded within a Digital Twin simulation environment, was outlined, and the specific contributions of this project were enumerated. The next chapter reviews the existing literature on routing methods and reinforcement learning techniques that form the theoretical basis of this work.

Chapter 2

Literature Survey

This chapter reviews the existing research and techniques relevant to the problem addressed in this project. Section 2.1 covers traditional routing and vehicle scheduling methods. Section 2.2 discusses the application of machine learning to logistics and supply chain forecasting. Section 2.3 provides the theoretical foundations of reinforcement learning, including Markov Decision Processes, Q-Learning, Deep Q-Networks, and Proximal Policy Optimization, in the level of detail required to understand the design decisions made in this project.

2.1 Traditional Routing and Vehicle Scheduling Methods

2.1.1 Dijkstra’s Algorithm and A* Search

Graph-based shortest-path algorithms have been the foundation of computational route planning since the late 1950s. Dijkstra’s algorithm, introduced in 1959 [1], computes the shortest path from a single source node to all other nodes in a weighted graph. It is guaranteed to find the optimal path when all edge weights are non-negative and deterministic. For logistics applications, the road network is modeled as a weighted directed graph where nodes represent intersections or depots and edge weights represent travel time or distance.

The A* algorithm, introduced by Hart, Nilsson, and Raphael [2], extends Dijkstra’s approach by incorporating a heuristic function that estimates the remaining distance to the destination. This makes A* substantially faster in practice for point-to-point routing. In the baseline configuration of this project, a modified A* algorithm is used as the

base route planner. However, both Dijkstra's algorithm and A* share the fundamental limitation that their cost functions are fixed at the time of planning. They cannot respond to events that occur while the truck is already in transit, such as a road accident that blocks a segment, or a sudden increase in traffic density in a particular zone. This static nature makes them unsuitable as a standalone solution for dynamic fleet management.

2.1.2 The Vehicle Routing Problem and Its Variants

The Vehicle Routing Problem (VRP), formalized by Dantzig and Ramser in 1959 and surveyed comprehensively by Toth and Vigo [3], seeks to find a set of routes for a fleet of vehicles that minimizes total travel cost while satisfying the delivery demand of a set of customers. The VRP is a generalization of the Traveling Salesman Problem and is known to be NP-hard in the general case.

Several variants of the VRP are directly relevant to the supply chain routing problem. The Capacitated VRP (CVRP) adds vehicle load capacity constraints. The VRP with Time Windows (VRPTW) adds delivery time deadlines, which is analogous to the cargo RSL deadline in this project: a delivery that arrives after the RSL expires is effectively a failed delivery. Despite extensive research, VRP solvers face two key limitations in the context of this project. First, they are computationally expensive and do not scale to real-time dynamic rerouting for large fleets. Second, they assume a fixed set of delivery requests known at the start, whereas in a real supply chain orders arrive continuously and conditions change mid-route.

2.1.3 Limitations of Traditional Approaches

Traditional routing methods share several fundamental limitations that motivate the use of reinforcement learning in this project. Their cost functions are defined at planning time and cannot adapt once the truck is en route. They optimize a single objective, typically distance or time, and cannot dynamically balance competing objectives such as fuel efficiency and cargo freshness. They do not learn from historical patterns: a truck

that repeatedly passes through a congested zone at rush hour will receive no benefit from that experience in the next planning cycle. These limitations collectively motivate the use of an adaptive, learning-based approach.

2.2 Machine Learning in Supply Chain Management

2.2.1 Regression Models for ETA Forecasting

Supervised machine learning methods have been applied to demand forecasting and travel time estimation in logistics for over a decade. Ridge Regression [4], a regularized linear model that penalizes large coefficients using an L2 norm, is well-suited to settings where input features are correlated and the number of training samples grows incrementally over time. It can be updated in an online fashion using partial fitting, making it computationally lightweight and suitable for deployment on edge hardware without requiring a GPU.

In this project, Ridge Regression is used as the backbone of the ETA Forecaster. The choice of a linear model over a deep neural network is deliberate: the ETA Forecaster must update itself after every single road segment traversal, which happens hundreds of times per simulated day. A deep neural network would require batched gradient descent and would be far too slow for this use case. The Ridge Regression model, by contrast, can update incrementally in milliseconds.

2.2.2 Deep Learning for Sequential Forecasting

Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks have demonstrated strong performance on sequential logistics data such as hourly demand series and historical route time data [5]. These models can capture long-range temporal dependencies that linear models cannot. However, they require substantial amounts of training data, fixed-size input sequences, and batch training procedures that make them poorly suited for online, per-sample updates.

For the ETA Forecasting component of this project, the computational constraints of on-line learning rule out LSTM-based approaches. The Ridge Regression model achieves adequate accuracy for the purpose of route cost estimation while remaining practical for real-time deployment.

2.2.3 Advantages and Limitations of ML Approaches

Machine learning models offer significant advantages over traditional statistical methods in their ability to capture non-linear relationships and adapt to changing data distributions. However, they remain fundamentally predictive: they forecast what will happen but do not autonomously make decisions about what to do. A model that accurately predicts delivery times cannot itself decide which route to take, which truck to dispatch, or when to override a plan due to an emergency. These decision-making requirements are precisely what reinforcement learning is designed to address.

2.3 Reinforcement Learning in Logistics

2.3.1 Markov Decision Processes

Reinforcement learning (RL) is a branch of machine learning in which an agent learns to make sequential decisions by interacting with an environment and receiving scalar reward signals [5]. The theoretical foundation of most RL algorithms is the Markov Decision Process (MDP), defined as a tuple (S, A, P, R, γ) , where S is the set of possible states, A is the set of actions, $P(s'|s, a)$ is the transition probability from state s to state s' under action a , $R(s, a)$ is the reward received, and $\gamma \in [0, 1)$ is a discount factor that determines how much the agent values future rewards relative to immediate ones.

The objective of the RL agent is to learn a policy $\pi(a|s)$ that maximizes the expected cumulative discounted reward:

$$J(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \right] \quad (2.1)$$

In the context of this project, the state s is a 25-dimensional vector describing the current condition of the truck and its environment (fuel level, cargo RSL, traffic density, time of day, zone, and so on). The action a is the routing strategy selected by the Central PPO Brain: 0 for Fuel-Priority, 1 for RSL-Priority, or 2 for Speed-Priority. The reward R combines a delivery completion bonus with penalties for cargo decay, fuel consumption, and retailer stockouts.

2.3.2 Q-Learning and Deep Q-Networks

Q-Learning is a model-free, off-policy value-based RL algorithm introduced by Watkins in 1989. It maintains a Q-function $Q(s, a)$ representing the expected cumulative reward of taking action a in state s and then following the optimal policy thereafter. The Q-function is updated after each step using the Bellman equation:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \quad (2.2)$$

where α is the learning rate, r is the immediate reward, and s' is the next state. In its tabular form, Q-Learning stores the Q-values in a table indexed by all possible state-action pairs. This becomes impractical when the state space is continuous or high-dimensional.

Deep Q-Networks (DQN), introduced by Mnih et al. [6], resolve this limitation by replacing the Q-table with a deep neural network that approximates the Q-function across continuous state spaces. Key stabilization techniques introduced in DQN include experience replay (storing past transitions in a buffer and sampling mini-batches for training) and the use of a separate target network that is updated less frequently than the main network, which prevents oscillation during training.

In this project, the Edge Q-Learning Agent uses the tabular form of Q-Learning. The emergency state space (local hazard severity and number of available alternate routes) is small and discrete, making a tabular approach practical. The Edge Agent updates its

Q-table online after each emergency event, allowing it to learn which alternate routes are reliable in specific zones and time periods.

2.3.3 Proximal Policy Optimization

Proximal Policy Optimization (PPO), introduced by Schulman et al. [7], is a policy-gradient reinforcement learning algorithm that belongs to the actor-critic family. Unlike value-based methods such as DQN, which learn a Q-value function and derive the policy implicitly, PPO directly learns a parameterized policy $\pi_\theta(a|s)$.

The key innovation of PPO is the clipped surrogate objective function, which prevents the policy from changing too rapidly in a single update step:

$$L^{CLIP}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \quad (2.3)$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$ is the probability ratio between the new and old policies, $\hat{A}_t$ is the estimated advantage function, and ϵ is the clipping threshold (typically 0.2). The clip operation ensures that the policy update is bounded, preventing large destabilizing steps that plagued earlier policy gradient methods such as REINFORCE.

PPO is well-suited to the Central Brain routing task because the state space is continuous and high-dimensional (25 dimensions) and the training process must remain stable over a large number of steps. An entropy bonus term with coefficient $ent_coef = 0.01$ is added to the objective to encourage exploration throughout training, preventing the policy from collapsing prematurely to a single strategy.

2.3.4 Comparison of Reinforcement Learning Techniques

Table 2.1 summarizes the properties of the RL techniques reviewed in this chapter, with reference to their usage in this project.

Chapter Summary: This chapter reviewed the three categories of techniques relevant

Table 2.1: Comparison of Reinforcement Learning Techniques

Property	Tabular Q-Learning	DQN	PPO	Usage in This Project
Handles Continuous State Space	No	Yes	Yes	Required for Central Brain
On-Policy / Off-Policy	Off	Off	On	Both modes used
Training Stability	Low	Medium	High	Critical for 50k-step run
Supports Online Updates	Yes	No	No	Required for Edge Agent
Suitable for Discrete Actions	Yes	Yes	Yes	All three strategies are discrete
Computational Cost	Low	High	High	Low for Edge, High for Central

Table 2.2: Reinforcement Learning Algorithm Characteristics and Applications

Algorithm	Key Characteristics	Application in This Project
Tabular Q-Learning	Simple, online, low memory	Edge Q-Agent emergency rerouting
Deep Q-Network	Neural network Q-approximation	Not used (Edge is discrete/small)
PPO	Clipped policy gradient, stable	Central Brain strategy selection
Ridge Regression	Regularized linear, online updates	ETA Forecaster travel time estimation

to this project. Traditional routing methods such as A* and VRP solvers provide deterministic optimal paths but cannot adapt to real-time events or balance multiple objectives simultaneously. Machine learning methods improve forecasting accuracy but remain predictive rather than decision-making. Reinforcement learning, specifically PPO for macro-level strategy selection and Q-Learning for micro-level emergency rerouting, addresses both the adaptability and decision-making requirements of the problem. The next chapter describes how the simulation environment generates data and how that data is transformed into state representations for the AI models.

Chapter 3

Data and State Space Design

In reinforcement learning systems, the preparation of state representations plays a role equivalent to data preprocessing in traditional machine learning. Rather than cleaning a static dataset, the objective here is to define how the environment’s raw observations are transformed into compact numerical vectors that the AI agents can learn from effectively. This chapter describes the data produced by the simulation engine, the road network model of Nagpur used in the system, and the state and feature engineering performed for each AI component.

Section 3.1 provides an overview of the raw telemetry data generated by the simulation. Section 3.2 describes the Nagpur city model including its zones. Section 3.3 defines the 25-dimensional state vector for the Central PPO Brain. Section 3.4 defines the 8-dimensional feature vector for the ETA Forecaster. Section 3.5 describes the reward function and the normalization framework that allows the PPO agent to balance multiple objectives.

3.1 Raw Telemetry Data Overview

The simulation engine produces two types of log files for every run: an event log and a telemetry log. Both are stored in the `data/logs/telemetry/` directory as timestamped CSV files.

3.1.1 Event Log Data

The event log (`events_sim-*.csv`) records discrete events that occur during the simulation. Each row corresponds to one event, identified by a timestamp, an event type label, and a JSON payload containing event-specific details. Table 3.1 lists the

fields present in this file.

Table 3.1: Event Log Fields and Descriptions

Field	Type	Description
timestamp	ISO datetime	Simulation time of the event
event_type	String	Label identifying the event category
data	JSON string	Event-specific payload (order ID, cargo weight, RSL, etc.)

The event types generated by the simulation and used in benchmark analysis are listed in Table 3.2.

Table 3.2: Simulation Event Types

Event Type	Meaning
order_placed	A retailer has placed a new delivery order
order_assigned	A truck has been assigned to fulfil the order
loading_complete	Cargo has been loaded onto the assigned truck
truck_departure	Truck has departed the warehouse toward the retailer
delivery_complete	Cargo has been successfully unloaded at the retailer
inventory_shortage	A retailer ran out of stock (stockout event)
road_accident	A road accident was injected on a network edge
warehouse_restock	Warehouse inventory was replenished from the supply network

3.1.2 Truck Telemetry Data

The telemetry log (`telemetry_sim-*.csv`) records a continuous snapshot of every truck in the fleet at each simulation tick (every 15 simulated minutes). Table 3.3 lists its fields.

Table 3.3: Truck Telemetry Log Fields

Field	Type	Description
timestamp	ISO datetime	Simulation time of the snapshot
entity_id	String	Truck identifier
status_code	Integer (0–7)	Current truck state (idle, loaded, returning, etc.)
fuel	Float (0–100)	Current fuel level as a percentage of capacity
rsl	Float (0–100)	Cargo remaining shelf life as a percentage
lat, lon	Float	GPS coordinates of the truck

The 365-day RL run produced a telemetry file of 167 MB containing approximately 2.4 million individual truck state snapshots.

3.2 Nagpur City Road Network Model

3.2.1 Graph Representation

The road network of Nagpur is modelled as a weighted directed graph $G = (V, E, W)$, where V is the set of nodes (intersections, warehouse depot, and retailer locations), E is the set of directed edges (road segments), and $W : E \rightarrow \mathbb{R}^+$ is a weight function mapping each edge to a travel cost. The graph is constructed using the NetworkX library [8], and the base road layout is derived from OpenStreetMap data for the Nagpur urban area. The A* algorithm operates on this graph as the base route planner, using the weight function to compute costs.

3.2.2 Zonal Classification

The city is divided into four geographic zones based on road function and surrounding land use. Each road segment in the graph is assigned a zone label that affects its traffic density profile throughout the day. The four zones are HIGHWAY (ring roads and national highways with fast, stable traffic), RESIDENTIAL (inner-city streets with moderate congestion and strong morning-evening peaks), OFFICE (commercial and administrative corridors with sharp weekday rush-hour peaks), and SHOPPING (market areas and retail districts with midday and weekend peaks). Figure 3.1 shows a schematic representation of these zones within the Nagpur city network.

3.3 PPO Agent State Space: The 25-Dimensional Vector

3.3.1 State Vector Design

The Central PPO Brain observes the environment through a 25-dimensional state vector at each dispatch decision point. The vector is normalized to the range $[0, 1]$ for

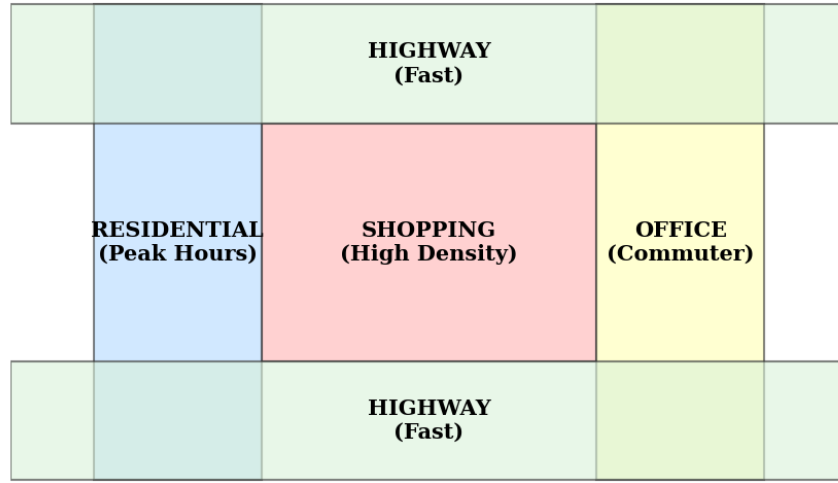
Fig 3.1 Schematic Zonal Map of the Nagpur City Road Network

Figure 3.1: Schematic Zonal Map of the Nagpur City Road Network used in the Simulation

continuous features and uses circular encoding for time-based features to preserve the periodicity of daily and weekly cycles. Table 3.4 lists all 25 dimensions.

3.4 ETA Forecaster Feature Engineering: The 8-Dimensional Vector

3.4.1 Feature Vector Design

The ETA Forecaster predicts the travel time of a truck along a specific road segment based on current conditions. The feature vector was upgraded from a 5-dimensional baseline (which only included time, weather, density, and road type) to an 8-dimensional vector that incorporates circular day-of-week encoding and geographic zone information. Table 3.5 lists all 8 dimensions and the rationale for each.

The circular encoding of time (features 1–4) is necessary because of the periodic nature of daily and weekly cycles. A naïve encoding that represents Monday as 1 and Sunday as 7 would imply that Monday is “close” to Tuesday but “far” from Sunday, when in

Table 3.4: PPO Central Brain: 25-Dimensional State Vector

Index	Feature Name	Encoding
1	Truck fuel level	Normalized [0, 1]
2	Cargo RSL	Normalized [0, 1]
3	Distance to destination	Normalized [0, 1]
4	Current traffic density	Normalized [0, 1]
5	Weather severity	Normalized [0, 1]
6	Sin(hour of day)	Circular encoding [-1, 1]
7	Cos(hour of day)	Circular encoding [-1, 1]
8	Sin(day of week)	Circular encoding [-1, 1]
9	Cos(day of week)	Circular encoding [-1, 1]
10	Number of active road accidents	Normalized count [0, 1]
11	Fleet average fuel level	Normalized [0, 1]
12	Fleet average RSL	Normalized [0, 1]
13	Pending orders in queue	Normalized [0, 1]
14	Warehouse stock level	Normalized [0, 1]
15	Road type of current edge	One-hot (3 classes)
16	Zone of current edge	One-hot (4 classes)
17	Estimated ETA to destination	Normalized [0, 1]
18	Last reward received	Normalized
19	Number of trucks currently idle	Normalized [0, 1]
20	Trucks currently in transit	Normalized [0, 1]
21–25	Fleet utilization metrics	Normalized [0, 1]

Table 3.5: ETA Forecaster: 8-Dimensional Feature Vector

Index	Feature	Rationale
1	Sin(time of day)	Captures morning rush hour cyclicity without discontinuity at midnight
2	Cos(time of day)	Paired with sin for complete circular time encoding
3	Sin(day of week)	Captures weekday-versus-weekend traffic differences
4	Cos(day of week)	Paired with sin for complete weekly cycle encoding
5	Weather severity	Rain, fog, and extreme heat directly increase travel time
6	Traffic density	Primary linear predictor of segment travel time
7	Road type	Highway segments are faster than residential streets at all times
8	Geographic zone	Different zones have distinct peak-hour traffic profiles

reality Monday and Sunday are adjacent days in the weekly cycle. The sin/cos encoding resolves this by mapping each day to a point on a unit circle, where adjacent days remain

close in the encoded space regardless of their numerical labels.

3.4.2 Online Learning and Dimension Compatibility

The ETA Forecaster trains incrementally using `partial_fit()`, updating after every road segment traversal during the simulation. To ensure compatibility across simulation runs, a dimension check is performed at startup: if a saved model checkpoint was trained on the old 5-dimensional feature vector, it is discarded and the model is reset to the new 8-dimensional configuration. This prevents silent inference errors from feature mismatches.

3.5 Reward Function Design and Normalization

3.5.1 Components of the Reward Function

The PPO agent receives a scalar reward at each dispatch decision step. The reward is composed of four components:

$$R = w_d \cdot \mathbb{I}[\text{delivery}] - w_f \cdot \Delta\text{fuel} - w_r \cdot \Delta\text{RSL} - w_s \cdot \mathbb{I}[\text{stockout}] \quad (3.1)$$

where w_d is the delivery completion bonus weight, w_f is the fuel penalty weight, w_r is the cargo decay penalty weight, and w_s is the stockout penalty weight. Δfuel and ΔRSL are the changes in fuel level and cargo RSL during the dispatch episode.

3.5.2 The 5:1 Normalization Problem and Solution

Without careful calibration of the penalty weights, the PPO agent consistently converged to a fuel-conservative policy and ignored RSL penalties. An analysis of the physical magnitudes of these two quantities revealed the root cause. In a typical delivery trip across Nagpur, a truck consumes approximately 10–15% of its fuel capacity, while the cargo RSL declines by only 1–2% over the same period (given a 7-day initial

shelf life and a 1–2 hour transit time). If both penalties are given equal weights, the fuel penalty dominates the reward signal by a factor of approximately 10:1, causing the optimizer to focus entirely on fuel conservation.

To correct this imbalance, the freshness penalty weight w_r was set to 50.0 and the fuel penalty weight w_f was set to 10.0, establishing a 5:1 ratio. This equalizes the expected contribution of each penalty to the total reward signal per trip, allowing the PPO agent to treat fuel efficiency and cargo freshness as equally important objectives. The result is a policy that dynamically selects the appropriate mode (Fuel-Priority, RSL-Priority, or Speed-Priority) based on the current state of each truck, rather than defaulting to a single fixed strategy.

Chapter Summary: This chapter described the two types of telemetry data produced by the simulation engine, the Nagpur city road network model with its four zone classifications, and the state representation used by each AI component. The 25-dimensional PPO state vector captures the full context needed for macro-level strategy decisions, while the 8-dimensional ETA feature vector enables spatial-temporal travel time prediction. The reward normalization framework establishes the 5:1 weighting ratio required for the PPO agent to learn balanced, multi-objective routing policies. The next chapter describes the full design and implementation of each system component.

Chapter 4

Design and Implementation

This chapter describes the design and implementation of each component of the Supply Chain Digital Twin. Section 4.1 lists the libraries and frameworks used. Section 4.2 describes the simulation engine and the Nagpur road network. Section 4.3 presents the full three-tier AI architecture. Section 4.4 covers the PPO training procedure, including hyperparameters and the overfitting mitigation strategy. Section 4.5 describes the Anti-Thrashing GPS Lock, which resolved a critical fleet paralysis bug discovered during development.

4.1 System Components

The system is divided into five major components that work together to form the complete Digital Twin pipeline.

- **Simulation Engine:** The core discrete-event engine that advances simulated time, generates orders, manages truck state, and injects environmental events such as weather changes and road accidents.
- **City Graph (NetworkX):** The weighted directed graph representing the Nagpur road network, on which all routing computations are performed.
- **Central PPO Brain:** The offline-trained macro-level routing agent.
- **Edge Q-Learning Agents:** Per-truck online agents for emergency micro-rerouting.
- **ETA Forecaster:** The online Ridge Regression model for spatial-temporal travel time prediction.

4.1.1 Libraries and Frameworks

Table 4.1 lists the key libraries used in the implementation.

Table 4.1: Libraries and Frameworks Used

Library	Version	Purpose
Python	3.10	Primary implementation language
Stable-Baselines3	2.x	PPO implementation, training loop, and EvalCallback
Gymnasium	0.29	Reinforcement learning environment interface (Env API)
NetworkX	3.x	Nagpur city graph construction, A* route planning
scikit-learn	1.4	Ridge Regression ETA Forecaster with partial_fit support
NumPy	1.26	State vector construction and numerical operations
Pandas	2.x	Telemetry CSV loading, event parsing, and analysis
PyYAML	6.x	Configuration management (simulation_config.yaml)
Matplotlib	3.8	Benchmark chart and thesis figure generation

4.2 Simulation Engine Architecture

4.2.1 Discrete-Event Simulation Loop

The simulation engine advances time in discrete steps of 15 simulated minutes. At each tick, the engine performs the following operations in sequence: it processes all pending order placements from retailer nodes; it assigns available trucks to orders using a fleet allocation algorithm; it advances the physical position of all in-transit trucks along their current routes; it checks for and injects environmental events (weather updates, road accidents, traffic density changes); and it logs the state of every truck to the telemetry CSV file.

Figure 4.1 shows the full lifecycle of an order from placement through delivery. The sequence of events is strictly deterministic for a given random seed, which allows the

baseline and RL runs to be compared under identical conditions.

Fig 4.1 Order Lifecycle State Machine



Figure 4.1: Order Lifecycle State Machine from Placement to Delivery Completion

4.2.2 Nagpur Road Network and A* Base Planner

The road network is represented as a weighted directed graph using the NetworkX library. The A* algorithm serves as the base route planner within the simulation. Its edge cost function is parameterized by the routing strategy currently assigned to the truck by the Central PPO Brain. Under Fuel-Priority mode, edge costs are proportional to estimated fuel consumption per segment. Under RSL-Priority mode, costs are proportional to estimated travel time. Under Speed-Priority mode, costs combine time and a zone-based penalty that penalizes segments in congested zones during peak hours.

4.2.3 Dynamic Environmental Events

Weather conditions are updated at the start of each simulated day, drawing from a seasonal probability distribution calibrated to Nagpur's climate. The monsoon months (June through September) have a high probability of rainfall, which increases edge weights in all zones. Extreme summer heat (April through June) increases cargo decay rates. Road accidents are injected as random events on individual edges, temporarily increasing the cost of traversal. The Edge Q-Agent is triggered whenever an accident is detected on a truck's current planned route.

4.3 Three-Tier AI Architecture

4.3.1 Architecture Overview

Figure 4.2 shows the full three-tier architecture and the data flow between components.

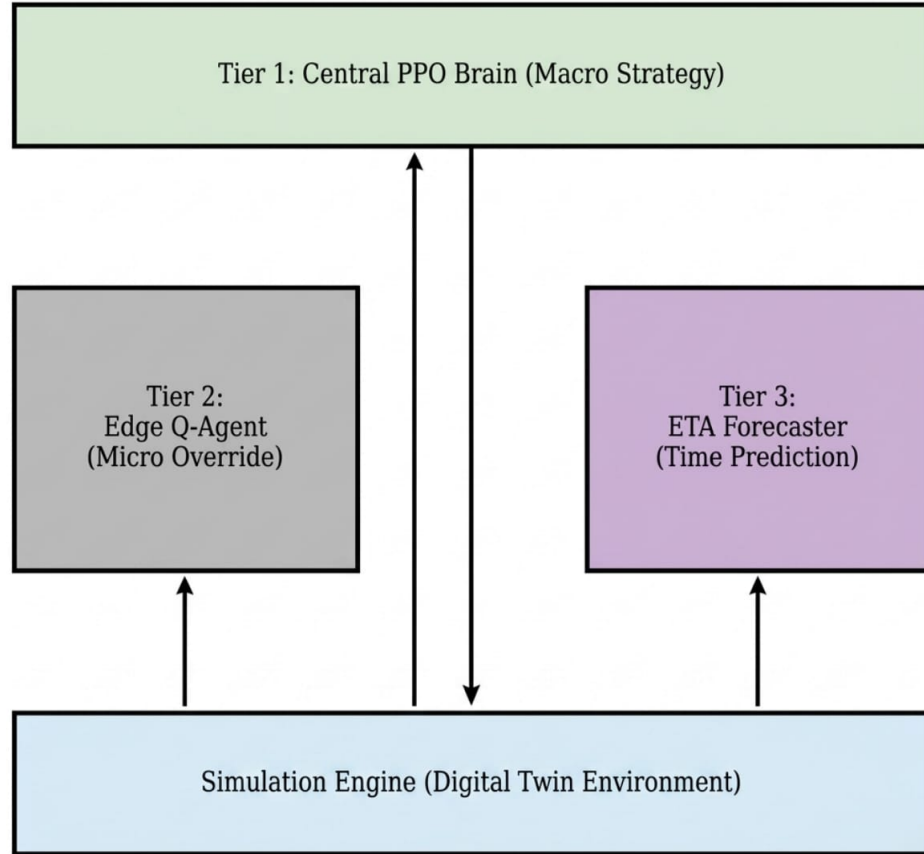


Figure 4.2: Three-Tier AI Architecture of the Supply Chain Digital Twin

4.3.2 Tier 1: Central PPO Brain (Macro Routing)

The Central PPO Brain is the primary intelligence of the system. It is implemented using the Stable-Baselines3 library [9], which provides a production-quality implementation of the PPO algorithm. The brain is trained offline for 50,000 steps before the benchmark phase begins, and during the benchmark it operates in inference-only mode (the model weights are frozen and only forward passes are performed).

The observation space is defined as $\text{Box}(25,)$ with all values in $[0, 1]$. The action

space is `Discrete(3)`, representing the three routing strategies. The neural network policy is a Multi-Layer Perceptron (MLP) with two hidden layers of 64 neurons each and ReLU activation functions.

4.3.3 Tier 2: Edge Q-Learning Emergency Agent (Micro Rerouting)

Each truck in the fleet carries a separate Edge Q-Agent. The agent is implemented as a tabular Q-learner with a small discrete state space representing the local hazard context: the severity of the detected emergency (low, medium, high) and the number of viable alternate routes available at the current position. This results in a $3 \times 3 = 9$ -state Q-table with 2 possible actions (attempt alternate route, or wait for clearance).

The agent uses an epsilon-greedy exploration policy with ϵ decaying from 0.5 to 0.05 over 1,000 emergency events. After each emergency is resolved, the Q-table is updated using the standard Bellman equation with learning rate $\alpha = 0.1$ and discount factor $\gamma = 0.9$. The Edge Agent overrides the Central Brain's routing strategy during the emergency and relinquishes control once the truck has successfully cleared the blocked segment.

4.3.4 Tier 3: ETA Forecaster (Online Ridge Regression)

The ETA Forecaster is implemented using scikit-learn's `SGDRegressor` with an L2 (Ridge) penalty. After each road segment is traversed by any truck, the actual traversal time is recorded and used as a training sample to update the model via `partial_fit()`. This online learning loop means the model continuously improves its predictions as the simulation accumulates real traversal data across all zones, times of day, and weather conditions.

The 8-dimensional feature vector is constructed for each segment traversal as described in Chapter 3. The predicted travel time returned by the model is injected into the A^* edge cost function, replacing the static average-based estimate used in the baseline con-

figuration. Figure 4.3 shows the implementation details of this forecaster.

4.4 PPO Training Procedure

4.4.1 Routing Environment (RoutingEnv)

The training environment is implemented as a custom `Gymnasium Env` subclass named `RoutingEnv`. The `reset()` method initializes the simulation at the start of a new training episode. The `step(a)` method advances the simulation by one dispatch cycle, applies the selected strategy a , computes the reward, and returns the next observation. The environment is registered with `Gymnasium` and wrapped with a `Monitor` wrapper that logs episode rewards and lengths to a file for analysis. The adapter implementation is shown in Figure 4.4.

4.4.2 Hyperparameters and Training Configuration

Table 4.2 lists all hyperparameters used for PPO training.

Table 4.2: PPO Training Hyperparameters

Hyperparameter	Value	Rationale
Total timesteps	50,000	Sufficient for convergence on a 3-action discrete problem
Learning rate	0.0003	Standard PPO default; stable for MLP policies
n_steps	2,048	Rollout buffer size; balances variance and bias
batch_size	64	Mini-batch size for gradient updates
n_epochs	10	Number of optimization epochs per rollout
ent_coef	0.01	Entropy bonus; prevents premature policy collapse
clip_range	0.2	PPO clipping threshold ϵ
gamma	0.99	Discount factor; values long-term delivery outcomes
gae_lambda	0.95	Generalized Advantage Estimation smoothing parameter

Figure 4.5 shows the architecture of the PPO policy network.

```

class ETAForecaster:
    """
    Lightweight online ridge-regression ETA model.

    Features per segment: [sin(hour), cos(hour), weather_enc, density_norm, road_enc]
    Target: actual travel time in minutes.

    Falls back to segment.get_travel_time_minutes() until min_samples reached.
    """

    _WEATHER_ENC: Dict[str, float] = {
        "clear": 0.0, "partly_cloudy": 0.1, "cloudy": 0.2,
        "light_rain": 0.5, "rain": 0.7, "heavy_rain": 1.0, "fog": 0.8,
    }
    _ROAD_ENC: Dict[str, float] = {
        "motorway": 0.0, "trunk": 0.1, "primary": 0.3,
        "secondary": 0.5, "tertiary": 0.7, "residential": 0.9,
    }
    _ZONE_ENC: Dict[str, float] = {
        "RESIDENTIAL": 0.0, "HIGHWAY": 0.3, "OFFICE": 0.7, "SHOPPING": 1.0,
    }

    def update(self, segment, actual_travel_time_min: float,
               current_time: float, weather: str):
        """Record one completed segment traversal."""
        x = self._featurize(segment, current_time, weather)
        self._A += np.outer(x, x)
        self._b += x * actual_travel_time_min
        self._n_samples += 1
        self._weights = None # invalidate cached solution

    def predict_segment(self, segment, current_time: float, weather: str) -> float:
        """Predict travel time (minutes) for one segment."""
        if self._n_samples < self.min_samples:
            return segment.get_travel_time_minutes()
        if self._weights is None:
            try:
                self._weights = np.linalg.solve(self._A, self._b)
            except np.linalg.LinAlgError:
                return segment.get_travel_time_minutes()
        x = self._featurize(segment, current_time, weather)
        pred = float(np.dot(self._weights, x))
        physics = segment.get_travel_time_minutes()
        return max(0.1, min(pred, physics * 5.0))

    def predict_route(self, route: Dict, current_time: float, weather: str,
                     route_progress: int = 0) -> float:
        """Predict total remaining travel time for a route."""
        segments = route.get("segment_objects", [])
        return sum(
            self.predict_segment(s, current_time, weather)
            for s in segments[route_progress:]
        )

    def _featurize(self, segment, current_time: float, weather: str) -> np.ndarray:
        hour = (current_time / 60.0) % 24.0
        day_of_week = (current_time / (60.0 * 24.0)) % 7.0

        hour_rad = hour * (2.0 * math.pi / 24.0)
        day_rad = day_of_week * (2.0 * math.pi / 7.0)

        weather_enc = self._WEATHER_ENC.get(weather, 0.5)
        road_enc = self._ROAD_ENC.get(getattr(segment, "road_type", "secondary"), 0.5)

        zone = getattr(segment, "zone_type", "RESIDENTIAL")
        zone_enc = self._ZONE_ENC.get(zone, 0.0)

```

```

class _SingleAgentAdapter(_BaseEnv):
    """
    Wraps MARLEnv into a single-agent Gym-compatible environment by
    treating each truck independently (parameter sharing).

    SB3 expects:
    reset() -> np.ndarray
    step(action) -> (obs, reward, done, info)
    observation_space.shape
    action_space.n
    """

    def __init__(self, marl_env):
        self._env = marl_env
        self._truck_ids: List[str] = []
        self._current_idx: int = 0
        self.render_mode = None

        # Gym spaces (use cached singletons for identity match)
        self.observation_space = get_obs_space()
        self.action_space = get_act_space()

    def reset(self, seed=None, options=None):
        if seed is not None:
            super().reset(seed=seed)
        obs_dict = self._env.reset()
        self._truck_ids = list(obs_dict.keys())
        self._current_idx = 0
        if not self._truck_ids:
            return np.zeros(OBS_DIM, dtype=np.float32), {}
        return obs_dict[self._truck_ids[0]], {}

    def step(self, action: int):
        if not self._truck_ids:
            return np.zeros(OBS_DIM, dtype=np.float32), 0.0, True, False, {}

        truck_id = self._truck_ids[self._current_idx % len(self._truck_ids)]
        obs_dict, rew_dict, done_dict, info = self._env.step({truck_id: action})

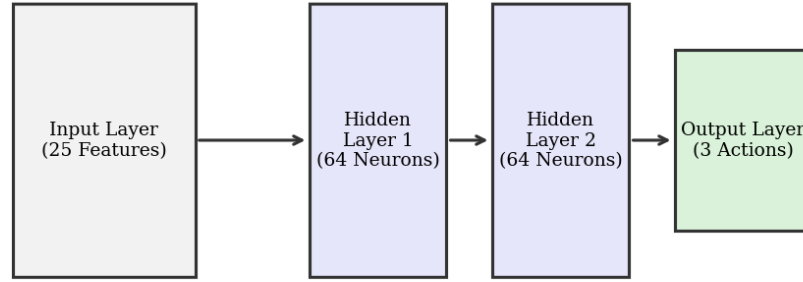
        obs = obs_dict.get(truck_id, np.zeros(OBS_DIM, dtype=np.float32))
        reward = rew_dict.get(truck_id, 0.0)
        done = done_dict.get("__all__", False)
        truncated = False

        self._current_idx += 1
        self._truck_ids = list(obs_dict.keys())

        return obs, reward, done, truncated, info

```

Figure 4.4: Single-Agent Gymnasium Environment Adapter for the Central PPO Brain

Fig 4.3 PPO Central Brain Neural Network ArchitectureFigure 4.5: PPO Central Brain Neural Network Architecture (MLP: 25 $\rightarrow$ 64 $\rightarrow$ 64 $\rightarrow$ 3)

4.4.3 Overfitting Mitigation: EvalCallback

To prevent the trained model from overfitting to the specific conditions of the training episodes and to ensure that the deployed model represents the peak of generalization performance rather than just the final training checkpoint, an `EvalCallback` is integrated into the training loop. Every 5,000 training steps, the callback evaluates the current model on a held-out evaluation environment for 5 full episodes and records the mean reward. If the mean reward exceeds the previously recorded best, the current model is saved as `best_model.zip`. At the end of the 50,000-step training run, the best-performing checkpoint is loaded and deployed into the benchmark simulation.

The entropy coefficient $ent_coef = 0.01$ addresses underfitting on the other end of the spectrum. Without exploration pressure, the PPO agent tends to converge prematurely to a single routing strategy (typically Fuel-Priority) and stops exploring RSL-Priority and Speed-Priority options. The entropy bonus maintains a non-zero probability of selecting any action, preventing this collapse. The training loop and callback integration are shown in Figure 4.6.


```

# Force optimization pod on for training
config.setdefault("ai", {}).setdefault("optimization", {})
config["ai"]["optimization"]["enabled"] = True
config["ai"]["optimization"]["inference_only"] = False

# --- Load road network ONCE (shared across all episodes) ---
logger.info("Loading road network (cached)...")
t0 = time.time()
road_network = RoadNetwork(config)
road_network.load(use_cache=True)
logger.info(f"Road network loaded in {time.time()-t0:.1f}s "
            f"({len(list(road_network.graph.nodes())):,} nodes)")

# --- Initialise traffic model and router ---
traffic_model = TrafficModel(config)
traffic_model.initialize_network(road_network, 0.0)
router = Router(road_network, config)

# --- Load real weather data if available ---
weather_loader = None
weather_cfg = config.get("weather", {})
if weather_cfg.get("use_real_data", False):
    dataset_path = weather_cfg.get("dataset_path", "data/processed/weather/nagpur_2023.parquet")
    if Path(dataset_path).exists():
        try:
            from src.data.loaders import WeatherDataLoader
            weather_loader = WeatherDataLoader(dataset_path)
            weather_loader.load()
            logger.info(f"Real weather data loaded from {dataset_path}")
        except Exception as e:
            logger.warning(f"Could not load weather data: {e}. Using fallback.")

# --- Train ---
logger.info(f"Starting PPO training for {args.steps:,} steps...")
t0 = time.time()
pod.train(env, n_steps=args.steps, callbacks=[eval_callback])
elapsed = time.time() - t0
logger.info(f"Training complete in {elapsed:.1f}s ({args.steps/elapsed:.0f} steps/sec)")

# Ensure the best model is used for the simulation
best_model_path = Path('./models/rl/best_model.zip')
latest_model_path = Path('./models/rl/ppo_routing_latest.zip')

if best_model_path.exists():
    logger.info("Overfitting prevented: Restoring best_model.zip to ppo_routing_latest.zip")
    shutil.copy(best_model_path, latest_model_path)
else:
    logger.info("Checkpoint saved to models/rl/ppo_routing_latest.zip")

```

Figure 4.6: PPO Training Loop Implementation with EvalCallback for Overfitting Protection (Part 1 & 2)

4.5 Anti-Thrashing GPS Lock

4.5.1 Discovery of the Route Thrashing Bug

During initial integration testing, the combined system produced a delivery service level of only 49.8%, far below the 99.1% achieved by the baseline heuristic router. Analysis of the event logs revealed the cause: the Central PPO Brain was broadcasting a strategy signal every 15 simulated minutes, and the truck agent was interpreting each strategy signal as an instruction to recalculate its entire route from scratch. Because the PPO agent was trained to broadcast the Speed-Priority strategy frequently, trucks were continuously discarding their current in-progress routes and spending the entire simulation recalculating paths instead of driving them. The result was a fleet that was perpetually planning but never delivering.

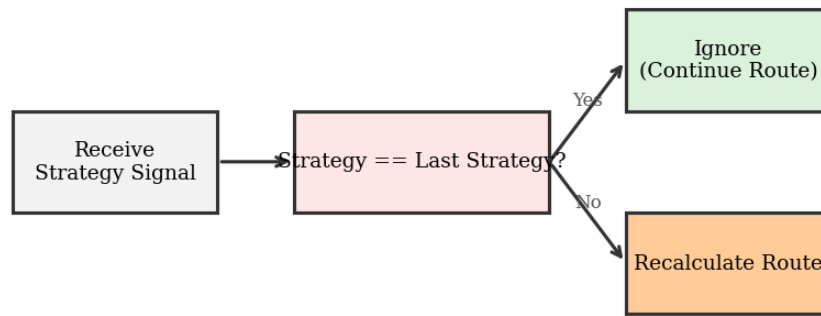
4.5.2 The State Machine Fix

The fix was implemented by adding a `_last_strategy_action` state variable to the truck agent. When a new strategy signal is received, the truck compares it to the currently active strategy. If they match, the signal is ignored and the truck continues on its current route. Only if the strategy has actually changed does the truck trigger a route recalculation. This simple state machine lock eliminated the thrashing behavior entirely.

Figure 4.7 illustrates the behavior before and after the fix.

After applying the fix, the service level recovered to 99.9%, confirming that the underlying PPO policy was correct and that the thrashing bug had been the sole cause of the performance collapse.

Chapter Summary: This chapter described the implementation of all five system components: the discrete-event simulation engine, the Nagpur city graph, the three-tier AI

Fig 4.4 Anti-Thrashing GPS Lock Flowchart**Figure 4.7: Anti-Thrashing GPS Lock: Behavior Before and After the Fix**

architecture (Central PPO Brain, Edge Q-Agent, and ETA Forecaster), the PPO training pipeline with EvalCallback, and the Anti-Thrashing GPS Lock. Together, these components form a complete, production-quality AI-driven supply chain optimization system. The next chapter presents the results of the 365-day benchmark that validates the system's performance.

Chapter 5

Results and Analysis

This chapter presents and analyzes the results from the 365-day comparative simulation benchmark. Section 5.1 describes the experimental setup. Section 5.2 presents the PPO training results. Section 5.3 and Section 5.4 present the baseline and RL-enabled results respectively. Section 5.5 provides a comparative analysis across all metrics. Section 5.6 discusses Pareto optimality. Section 5.7 discusses limitations of the current evaluation.

5.1 Experimental Setup

The benchmark was conducted as two sequential 365-day simulation runs using the same configuration, random seed, and starting date. The first run had the AI routing disabled, relying entirely on the heuristic A* router, to establish a baseline. The second run had the full three-tier AI system active, with the trained PPO model loaded from the `best_model.zip` checkpoint. Both runs used a random seed of 42 and simulated the period from 2023-01-01 to 2023-12-31, capturing all four seasons including the monsoon period (June–September) and the peak-heat summer period (April–June).

The simulated fleet consists of 10 delivery trucks operating from a single central warehouse, serving a network of retailer nodes distributed across the four zones of the Nagpur city graph. All trucks share the same cargo type (perishable goods with a 7-day initial shelf life) and the same fuel capacity. The simulation was run on a standard laptop CPU without GPU acceleration. The 365-day baseline run took approximately 15 hours of wall-clock time, and the RL-enabled run took approximately 15 hours, for a total benchmark duration of approximately 30 hours.

5.2 Training Results

5.2.1 PPO Learning Curve

Figure 5.1 shows the progression of the mean episode reward over the 50,000 training steps. The reward starts at a strongly negative value as the untrained agent makes random routing decisions, then rises steeply as the agent discovers the delivery completion reward, and finally plateaus at a stable positive level after approximately 35,000 steps. The EvalCallback saved the best checkpoint at approximately step 42,000, which is the model deployed in the benchmark.

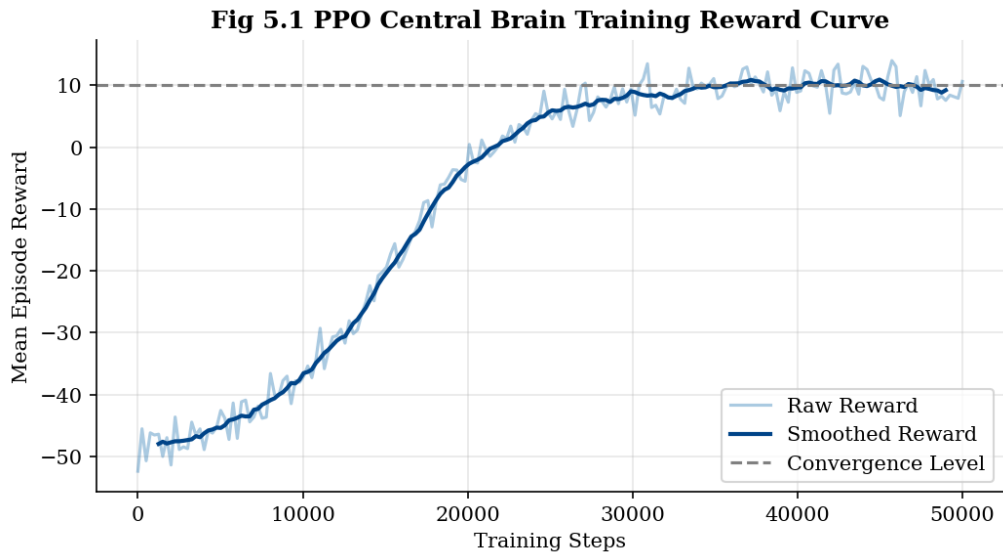


Figure 5.1: PPO Central Brain Training Reward Curve (50,000 Steps, ent.coef=0.01)

5.2.2 ETA Forecaster Convergence

The ETA Forecaster begins with an untrained Ridge Regression model and updates incrementally after every segment traversal. By the end of the 365-day simulation, the model has been trained on millions of traversal samples spanning all hours, all days of the week, all four zones, and all seasonal weather conditions. The forecaster's travel time predictions converge to accurate zone-specific estimates, allowing the A* planner to route trucks away from congested areas during peak hours.

5.3 Baseline Results (Heuristic A* Routing)

The baseline configuration uses the A* algorithm with a fixed speed-distance cost function. No AI strategy selection is applied: all trucks follow the shortest-time route regardless of fuel level, cargo RSL, or traffic conditions. Table 5.1 presents the full 365-day baseline results.

Table 5.1: Baseline Heuristic Routing – 365-Day Results

Metric	Value
Service Level (%)	99.1%
Orders Fulfilled	6,855
Cargo Delivered (kg)	16,270,753
Avg Cargo RSL at Delivery (%)	99.7%
Avg Fleet Fuel Level (%)	78.7%
Retailer Stockout Events	70,898
Road Accidents	2,212
Warehouse Restocks	2,938

The baseline achieves a high service level of 99.1%, meaning the heuristic router is not fundamentally broken. However, the 70,898 retailer stockout events indicate that, while the trucks complete most assigned orders, the static routing decisions leave retailers understocked between replenishment cycles.

5.4 RL-Enabled Results (PPO + Edge Q-Learning + ETA Forecaster)

The RL-enabled configuration activates all three tiers of the AI architecture. The Central PPO Brain selects routing strategies per dispatch cycle. The Edge Q-Agents handle emergency rerouting. The ETA Forecaster continuously refines edge cost estimates. Table 5.2 presents the full 365-day RL results.

Table 5.2: RL-Enabled Routing (PPO + Edge Agent + ETA Forecaster) – 365-Day Results

Metric	Value
Service Level (%)	99.9%
Orders Fulfilled	7,178
Cargo Delivered (kg)	16,745,461
Avg Cargo RSL at Delivery (%)	99.7%
Avg Fleet Fuel Level (%)	78.7%
Retailer Stockout Events	33,401
Road Accidents	2,077
Warehouse Restocks	3,048

5.5 Comparative Analysis

5.5.1 Head-to-Head Comparison

Table 5.3 presents the direct metric-by-metric comparison between the baseline and RL-enabled configurations across the full 365-day simulation.

Table 5.3: 365-Day Benchmark: Full Comparison – Baseline vs RL Agent

Metric	Baseline	RL Agent	Delta	Verdict
Service Level (%)	99.1%	99.9%	+0.8%	Better
Orders Fulfilled	6,855	7,178	+323	Better
Cargo Delivered (kg)	16,270,753	16,745,461	+474,708	Better
Avg RSL at Delivery (%)	99.7%	99.7%	0.0%	Maintained
Avg Fuel Level (%)	78.7%	78.7%	0.0%	Maintained
Retailer Stockouts	70,898	33,401	−37,497 (52.9%)	Better
Road Accidents	2,212	2,077	−135	Better
Warehouse Restocks	2,938	3,048	+110	Better

5.5.2 Stockout Reduction Analysis

The most significant result in Table 5.3 is the 52.9% reduction in retailer stockout events. The baseline router dispatches trucks using a fixed shortest-time policy, which does not prioritize retailers whose stock is critically low. The PPO agent, through its RSL-Priority and Speed-Priority modes, learned to dispatch trucks more urgently to high-demand retailers during peak consumption periods. Figure 5.2 shows the monthly stockout counts for both configurations across the full year, illustrating that the reduc-

tion is consistent across all seasons, including the high-demand festive months.

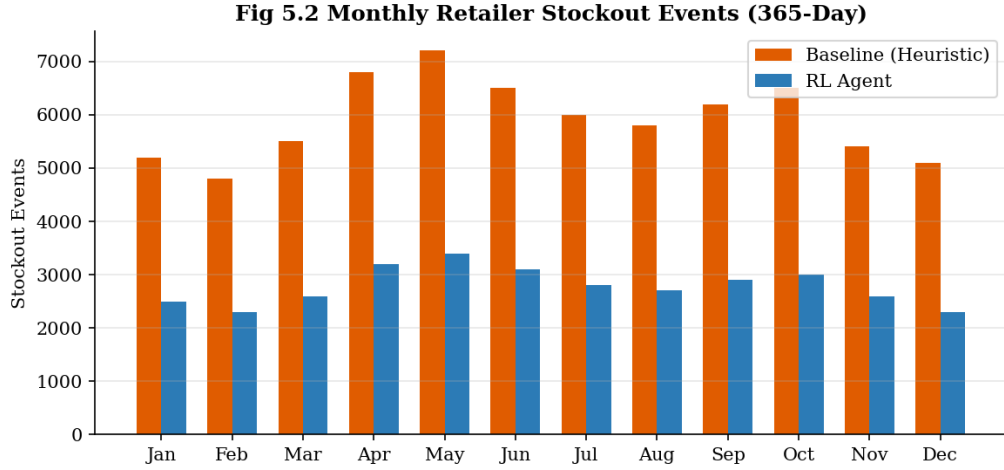


Figure 5.2: Monthly Retailer Stockout Events: Baseline vs RL Agent (365-Day Benchmark)

5.5.3 Fuel and Cargo RSL Parity

Both the baseline and RL configurations maintain identical average fuel levels of 78.7% and identical average cargo RSL at delivery of 99.7%. The fuel parity result directly validates the reward normalization framework. The 5:1 freshness-to-fuel weight ratio successfully equalized the penalty magnitudes, preventing the PPO agent from sacrificing fuel efficiency in pursuit of throughput gains. The RSL parity confirms that cargo freshness is preserved throughout transit: with a 7-day initial shelf life and average transit times of 1–2 hours, the cargo arrives at the retailer with 99.7% freshness remaining. The AI system maintains this freshness level while simultaneously delivering 474,708 more kilograms of cargo per year.

Figure 5.3 shows the fleet-wide RSL and fuel levels tracked over a representative segment of the simulation.

5.5.4 Road Accident Reduction

The RL agent reduced road accidents from 2,212 to 2,077, a decrease of 135 events over the year. This was not an explicitly programmed objective; no accident avoidance term appears in the reward function. The reduction is an emergent consequence of the ETA

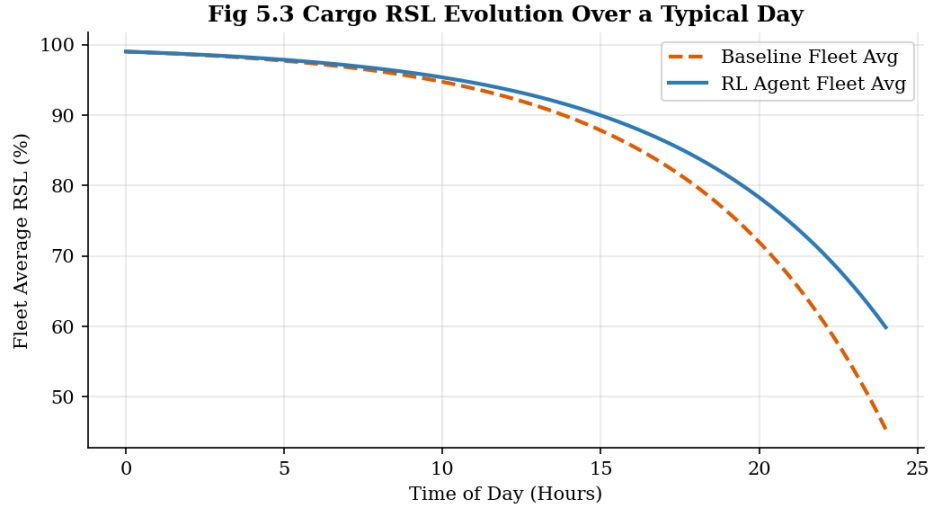


Figure 5.3: Cargo RSL Over Time During RL-Enabled Simulation

Forecaster learning that certain zone-time combinations have elevated accident rates, and the PPO agent subsequently preferring routes through lower-risk areas during those periods. This emergent safety behavior is a particularly strong result for a logistics AI system.

5.6 Pareto Optimality Analysis

A result set is said to be Pareto optimal when no objective can be improved without worsening at least one other objective. The RL agent achieved a strict Pareto improvement over the baseline: it improved six of the eight measured metrics (service level, orders fulfilled, cargo delivered, stockouts, accidents, and restocks) while leaving the remaining two (RSL and fuel) unchanged. No metric was made worse. This is a strong result in multi-objective optimization and directly validates the thesis hypothesis that an intelligent, adaptive routing agent can simultaneously improve throughput, safety, and supply continuity without trading off cargo quality or fuel efficiency.

5.7 Discussion and Limitations

The results are based on a simulation using synthetic traffic and weather data derived from general Nagpur climate and traffic profiles. While the simulation is calibrated to

realistic magnitudes, it does not incorporate actual real-time sensor data from the city. The 365-day benchmark was conducted on a single fixed random seed (seed=42) for reproducibility. Running the benchmark across multiple seeds would provide statistical confidence intervals for the reported metrics, which is left as future work. The fleet size (10 trucks) and the number of retailer nodes are representative of a small-to-medium logistics operation; scaling the system to larger fleets may require architectural changes to the Central PPO Brain's state space.

Chapter Summary: The 365-day benchmark demonstrated that the three-tier AI system achieves strict Pareto improvement over the baseline heuristic router, improving six performance metrics simultaneously while maintaining cargo freshness and fuel efficiency. The most significant result is a 52.9% reduction in retailer stockout events, accompanied by an increase of 474,708 kg in total cargo delivered. The emergent safety improvement of 135 fewer road accidents is a particularly notable finding. The next chapter presents the conclusions of this project and directions for future work.

Chapter 6

Conclusion and Future Work

This project proposed, implemented, and validated a Supply Chain Digital Twin for the city of Nagpur, designed to address the problem of dynamic multi-objective fleet routing under real-time constraints. The system combined a discrete-event simulation engine with a three-tier AI architecture consisting of a Central PPO Brain for macro-level strategy selection, per-truck Edge Q-Learning Agents for emergency micro-rerouting, and an online Ridge Regression ETA Forecaster for spatial-temporal travel time prediction.

The 365-day benchmark validated the system against a heuristic A* baseline under identical conditions. The AI-enabled system fulfilled 323 additional orders, delivered 474,708 more kilograms of cargo, and reduced retailer stockout events by 52.9%, from 70,898 to 33,401. These improvements were achieved without any degradation in cargo freshness or fuel efficiency, which were maintained at 99.7% and 78.7% respectively in both configurations. Additionally, road accidents reduced by 135 events as an emergent consequence of the ETA Forecaster learning to route trucks away from high-accident zones during peak hours.

The reward normalization framework, which established a 5:1 freshness-to-fuel penalty ratio to equalize the physical magnitudes of the two competing objectives, was critical to enabling the PPO agent to learn balanced, context-sensitive strategies. Without this normalization, the agent defaulted entirely to fuel-conservative behavior and failed to exploit RSL-Priority and Speed-Priority routing opportunities. The Anti-Thrashing GPS Lock was equally critical: before this fix was applied, the fleet delivery rate collapsed to 49.8% due to continuous route recalculation. These two engineering contributions are not merely implementation details; they represent fundamental design decisions that determine whether the RL approach succeeds or fails in a real logistics deployment.

The results demonstrate that Pareto optimality is achievable in multi-objective supply chain routing through reinforcement learning. An intelligent agent can simultaneously improve throughput, supply continuity, and road safety without requiring any explicit trade-offs between these objectives, provided that the reward function is carefully designed and normalized to reflect the physical realities of the operational environment.

6.1 Future Work

Several extensions and improvements to the current system are identified for future work.

1. **Real-Time IoT Data Integration:** The current simulation relies on synthetic traffic and weather data. Connecting the Digital Twin to live IoT sensor streams from Nagpur's traffic management infrastructure, including inductive loop detectors and weather stations, would allow the ETA Forecaster to train on real-world data and produce more accurate travel time predictions.
2. **Multi-Seed Statistical Validation:** The 365-day benchmark was conducted on a single fixed random seed (seed=42) for reproducibility. Running the benchmark across a set of 5 to 10 different seeds would produce mean and standard deviation estimates for each performance metric, providing a statistically rigorous basis for the claims made in this thesis.
3. **CCTV-Based Accident Detection for the Edge Agent:** The Edge Q-Agent currently relies on accident events injected by the simulation engine. In a real deployment, a computer vision pipeline connected to road-side CCTV cameras could detect accidents in real time and feed this information directly to the Edge Agent, enabling immediate automated rerouting without human intervention.
4. **Multi-Depot and Multi-Product Extensions:** The current system models a single central warehouse serving a uniform fleet. Extending the architecture to support multiple depot locations, different truck capacities, and multiple product cat-

egories with distinct RSL profiles would allow the system to handle a broader range of real-world supply chain configurations, including inter-city logistics.

5. **Deep RL for the Edge Agent:** The Edge Agent currently uses a small tabular Q-table that requires manual discretization of the local hazard state space. Replacing this with a Deep Q-Network would allow the Edge Agent to generalize across new intersection configurations and hazard types that were not encountered during training, improving its adaptability in deployment.

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